

Zero-DTE Gamma Scalping - MSTR & COIN Performance Report

WithAI Research

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Zero-DTE Gamma Scalping: MSTR & COIN Comparative Performance Report

Strategy: Long ATM Straddle + Delta-Neutral Hedging (Friday weekly 0-DTE)

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Summary Comparison

Metric	SPY	MSTR	COIN
Period	Mar 2024 - Apr 2026	Aug 2024 - Apr 2026	Mar 2024 - Apr 2026
Trading Days	343	36	30
Total P&L;	-\$8,527	-\$2,458	+\$2,383
Avg Daily P&L;	-\$24.86	-\$68.28	+\$79.42
Median Daily P&L;	-\$35.55	-\$18.37	+\$242.83
Std Dev	\$112.67	\$383.91	\$797.23
Sharpe Ratio	-3.50	-2.82	+1.58
Win Rate	25.95%	44.44%	70.00%
Best Day	+\$843.95	+\$993.21	+\$1,405.89
Worst Day	-\$428.14	-\$724.41	-\$2,379.34

Key Finding: COIN Was Profitable

COIN generated **+\$2,383 total P&L**; with a **70% win rate** and **1.58 Sharpe ratio** -- the only profitable configuration tested. This validates the core thesis: gamma scalping works when realized volatility exceeds implied volatility, which happens more frequently in high-vol crypto-correlated names.

MSTR Results (Aug 2024 - Apr 2026)

Period: Post 10:1 stock split (Aug 8, 2024) to present. Fridays only.

Metric	Value
Total P&L;	-\$2,457.99
Trading Days	36 / 87 Fridays (41.4% data coverage)
Avg Daily P&L;	-\$68.28

Metric	Value
Median Daily P&L;	-\$18.37
Win Rate	44.44% (16W / 20L)
Sharpe Ratio	-2.82

Win/Loss Payoff:

- Average winning day: ~\$275
- Average losing day: ~\$315
- Payoff ratio roughly symmetric but skewed slightly negative

Notable days:

- Best: +\$993.21 (Jun 20, 2025)
- Worst: -\$724.41 (Mar 14, 2025)

Why MSTR underperformed:

- MSTR trended strongly in both directions (up to \$450 then back to \$130)
- Despite high IV, persistent directional moves hurt delta-neutral strategy
- 44% win rate is much better than SPY (26%) but insufficient given cost structure
- Only 36 usable days out of 87 Fridays -- sparse option data limits sample size

COIN Results (Mar 2024 - Apr 2026)

Metric	Value
Total P&L;	+\$2,382.69
Trading Days	30 / 110 Fridays (27.3% data coverage)
Avg Daily P&L;	+\$79.42
Median Daily P&L;	+\$242.83
Win Rate	70.00% (21W / 9L)
Sharpe Ratio	+1.58

Win/Loss Payoff:

- Average winning day: ~\$363
- Average losing day: ~\$585
- Losses are larger per event, but far less frequent

Notable days:

- Best: +\$1,405.89 (May 16, 2025)
- Worst: -\$2,379.34 (Oct 10, 2025)

- Second worst: -\$2,306.76 (Aug 1, 2025)

Why COIN worked:

- High intraday volatility driven by crypto market correlation
- More mean-reverting price action on weekly expiry days
- Realized volatility frequently exceeded implied volatility
- 70% win rate overcomes the unfavorable payoff asymmetry (wins < losses)

Data Coverage Issue

A significant limitation: Alpaca minute-level option data coverage is sparse for individual stocks.

Symbol	Fridays Tested	Successful	Coverage
MSTR	87	36	41.4%
COIN	110	30	27.3%
SPY	546 weekdays	343	62.8%

Primary failure reason: Insufficient option bar count. Many days had call or put data below the 50-bar minimum, or merged data was too sparse. This is a data availability issue, not a strategy issue -- results should be interpreted with caution given the small sample sizes.

Risk Analysis

Tail Risk

COIN has significant tail risk: two days lost >\$2,300, which is 30x the average daily P&L.; Without those two outliers, COIN performance would be dramatically better:

Metric	With Outliers	Without 2 Worst Days
Total P&L;	+\$2,383	+\$7,069
Avg Daily P&L;	+\$79	+\$252
Win Rate	70%	75%

This suggests a **stop-loss at ~-\$1,000** would materially improve risk-adjusted returns.

Volatility Scaling

Payoff magnitude scales with underlying price and volatility:

Symbol	Avg Price	Daily Std Dev	Relative Scale
SPY	~\$550	\$113	1.0x

Symbol	Avg Price	Daily Std Dev	Relative Scale
MSTR	~\$250	\$384	3.4x
COIN	~\$250	\$797	7.1x

Higher per-trade variance in MSTR/COIN is expected given their higher underlying volatility.

Execution Parameters Used

Parameter	SPY	MSTR	COIN
Stock spread	\$0.01	\$0.05	\$0.05
Option spread	1.5%	2.5%	2.5%
Delta threshold	15 shares	15 shares	15 shares
Contracts	1	1	1
IV assumption	25%	25%	25%
0-DTE frequency	Daily	Fridays only	Fridays only

Recommendations

1. Scale COIN Strategy

COIN showed genuine edge. Next steps:

- Increase to 3-5 contracts to improve P&L; magnitude
- Add stop-loss at -\$1,000 to cap tail risk
- Calibrate IV from market straddle prices (25% constant is too low for COIN)

2. Regime Filter for MSTR

MSTR has the volatility but trends too aggressively. Add:

- Skip days when MSTR moved >5% in the prior week (trending signal)
- Only trade when BTC is range-bound (sideways crypto regime)

3. Data Quality

- Consider alternative data sources (OPRA feed, OptionMetrics) for better coverage
- The 27-41% coverage rate means >60% of potential trading days were unusable
- Higher coverage would increase sample size and statistical confidence

4. Parameter Optimization

- Test wider delta thresholds (20-30 shares) to reduce hedge costs on high-vol names

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- Test different entry times (10 AM vs 9:31 AM) -- individual stock options may have wider spreads at open
 - Use market-implied volatility instead of constant 25%
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Report Generated: April 5, 2026

Data Source: Alpaca API (minute-level historical options + stock data)

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